Gonzalo Benegas

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Summary

AI for Science researcher with a PhD in Computational Biology.

Experience

Open Athena, Research Scientist – New York, NY 2025 – present
- Developing training data and evaluations to advance scientific capabilities in Marin, an openly developed LLM.
- Led the development of MarinDNA, a 1B-parameter genomic language model competitive with Evo 2 40B on variant effect prediction while scoring variants 2,300× faster.

University of California, Berkeley, Postdoctoral Researcher, EECS – Berkeley, CA 2024 – 2025
- Co-developed GPN-Star, a phylogeny-informed genomic language model achieving SOTA variant effect prediction across both rare and common variants in the human genome (Nature, 2026).
- Co-authored a review on genomic language models, featured on the cover of Trends in Genetics (2025).
- Created TraitGym, a benchmark for evaluating DNA sequence models on variant effect prediction (bioRxiv, 2025). Used as evaluation benchmark by DeepMind's AlphaGenome (Nature, 2026).

University of California, Berkeley, Graduate Student Researcher – Berkeley, CA 2018 – 2023
- Developed GPN-MSA, the first alignment-based genomic language model, achieving SOTA variant effect prediction for rare variants and scalable inference across all 9B possible SNVs in the human genome (Nature Biotechnology, 2025).
- Developed GPN, the first genomic language model to exhibit zero-shot variant effect prediction capabilities across the entire genome (PNAS, 2023).
- Developed scQuint, a variational auto-encoder for analysis of alternative splicing in single-cell RNA-seq (eLife, 2022).

Autodesk, Software Engineer – Buenos Aires, Argentina 2015 – 2016

Synthicity, Software Engineer – Buenos Aires, Argentina 2014 – 2015

Education

PhD **University of California, Berkeley**, PhD in Computational Biology 2023
- Advisor: Yun S. Song

BS+MS **University of Buenos Aires**, BS+MS in Computer Science 2018
- GPA: 9.70/10

Selected Publications

Nature (Co-first author) 2026
Predicting genome-wide functional constraints with GPN-Star

Nature Biotechnology (First author) 2025
A DNA language model based on multispecies alignment predicts the effects of genome-wide variants

Trends in Genetics (Co-first author) 2025
Genomic Language Models: Opportunities and Challenges

bioRxiv (First author) 2025
Benchmarking DNA Sequence Models for Causal Regulatory Variant Prediction in Human Genetics

Proceedings of the National Academy of Sciences (First author) 2023
DNA language models are powerful predictors of genome-wide variant effects

eLife (First author) 2022
Robust and annotation-free analysis of alternative splicing across diverse cell types in mice